95.7
Pass

Ankita Gupta

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95.7% • 167.5 / 175
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Candidate Information

Email	mailankita.nimi@gmail.com
Test	eClerx - Java Coding Test - 3+ Years New
Candidate Packet	View
Taken on	13 Feb 2026 02:59:29 IST
Time taken	34 min 33 sec/ 90 min
Work Experience	> 5 years
Invited by	Kalyani
Invited on	12 Feb 2026 19:24:09 IST

Skill Distribution

No.	Skill	Score
1	Spring Boot Advanced	50%

2	Spring Boot Intermediate	83%
3	Problem Solving Intermediate	100%
4	Java Intermediate	100%

Tags Distribution

Spring Boot	70%	Hard	50%
Spring	50%	JPA	50%
Spring Data	50%	Transactional	50%
Medium	98%	Exception Handling	100%
Aspect-Oriented Programming	100%	Graphs	100%
Greedy	100%	Java	100%

Questions

Spring Boot (Advanced) • 17.5 / 25

No.	Question	Skill	Score	Code Quality	Time Taken	Status
1	Specification Multiple Choice	Spring Boot (Advanced)	5/5	-	39 sec	✓

2	Cracker Multiple Choice	Spring Boot (Advanced)	0/5	-	1 min 3 sec	✗
3	Spring Data Transactional Multiple Choice	Spring Boot (Intermediate)	2.5/5	-	1 min 6 sec	✓
4	Endpoint Security Multiple Choice	Spring Boot (Intermediate)	5/5	-	38 sec	✓
5	Predict the Output Multiple Choice	Spring Boot (Intermediate)	5/5	-	1 min 26 sec	✓

Problem Solving (Intermediate) • 75 / 75

No.	Question	Skill	Score	Code Quality	Time Taken	Status
6	Minimize Path Value Coding	Problem Solving (Intermediate)	75/75	-	21 min 32 sec	✓

Java (Intermediate) • 75 / 75

No.	Question	Skill	Score	Code Quality	Time Taken	Status
7	Java: Optimized Sequential Video Watching Coding	Java (Intermediate)	75/75	-	7 min 29 sec	✓

1. Specification

Correct

Multiple Choice | Spring Boot | Hard

Question description

Which of the following options can be used for dynamic jpa queries with spring-data-jpa repositories?

Candidate's Solution

Options: (Expected answer indicated with a tick)

- ○ PreparedStatement
- ● Specification ✓
- ○ Criteria
- ○ Query

No comments.

2. Cracker

Incorrect

Multiple Choice | Spring Boot | Hard

Question description

Cracker.java

```
@Component
public class Cracker {
    public void startCracking(){
        System.out.println("Cracking...");
    }
    public void stopCracking(){
        System.out.println("Cracking Stopped!");
    }
}
```

CrackerService.java

```
@Service
public class CrackerService {

    @Autowired
    Cracker cracker;
    public void workWithNewCracker() {
        cracker.stopCracking();
        cracker = new Cracker();
    }
    public void startWorking() {
        cracker.startCracking();
    }
}
```

CrackerWorker.java

```
@Service
public class CrackerWorker {

    @Autowired
    CrackerService crackerService;

    public void startWorking() {
        crackerService.startWorking();
        crackerService.workWithNewCracker();
        crackerService = new CrackerService();
        crackerService.startWorking();
    }
}
```

Consider these classes are managed in Spring Boot. What happens when 'crackerWorker.startWorking()' is called in the Spring context?

Interviewer guidelines

CrackerService in CrackerWorker will work perfectly until it's replaced with new CrackerService. When that happens it just replace existing injected bean with empty pojo. Which doesn't have it's cracker object so when it's startWorking method is called it will try to execute null cracker attributes startWorking method and NullPointerException will arise.

Candidate's Solution

Options: (Expected answer indicated with a tick)

- ○ "Cracking..." "Cracking Stopped!" will be written to console
- ● **"Cracking..." "Cracking Stopped!" "Cracking..." will be written to console**
- ○ "Cracking..." "Cracking Stopped!" NullPointerException will be thrown ✓
- ○ None of above

No comments.

3. Spring Data Transactional

Partially correct

Multiple Choice | Spring | Spring Boot | JPA | Spring Data | Transactional | Medium

Question description

Given the following code, what happens when *StudentRepsitoryTest* runs?

Lesson.java

```
@Entity
public class Lesson {
   @Id
   private Long id;
   private String name;
   private Integer courseLength;
   private Integer courseLevel;
   @JsonBackReference
   @ManyToOne(fetch = FetchType.LAZY)
   @JoinColumn(name = "student_id")
   private Student student;
   private boolean inProgress;
   //getter & setters
}
```

Student.java

```
@Entity
public class Student {
   @Id
   private Long id;
   private String name;
   private String surname;
   @JsonManagedReference
   @OneToMany(mappedBy = "student", fetch = FetchType.EAGER, cascade = {CascadeType.PERSIST,
CascadeType.REFRESH})
   private List<Lesson> lessons;
   //getter & setters
}
```

StudentRepository.java

```
@Repository
public interface StudentRepository extends JpaRepository<Student, Long>{
    //Y
    Optional<Student> findById(Long id);
}
```

LessonRepository.java

```
@Repository
public interface LessonRepository extends JpaRepository<Lesson, Long> {
}
```

StudentService.java

```
@Service
public class StudentService{
    private StudentRepository studentRepository;
    public StudentService(StudentRepository studentRepository, LessonRepository lessonRepository)
{
        this.studentRepository = studentRepository;
        this.lessonRepository = lessonRepository;
    }

    public void save(Student student) {
        studentRepository.save(student);
    }

    //other crud metods

    //X
    public void deleteStudentWithLessons(Long id) throws SQLException {
        Student student =
                studentRepository.findById(id).orElseThrow(() -> new RuntimeException("Student not foun
d"));
        List<Lesson> lessons = student.getLessons();
        lessons.stream().forEach(lesson -> lesson.setStudent(null));
        studentRepository.delete(student);

        boolean inProgress =lessons.stream()
                .map(Lesson::isInProgress)
```

```
                .filter(progress -> progress == true)
                .findFirst()
                .orElseGet(() -> false);
        if (inProgress) {
            throw new SQLException("Cannot delete all Lessons. There are lessons in progress state");
        }
        lessonRepository.deleteAll(lessons);
    }
}
```

StudentRepositoryTest.java

```
@SpringBootTest
class StudentRepositoryTest {
    @Autowired
    StudentRepository studentRepository;
    @Autowired
    StudentService studentService;
    @Autowired
    LessonRepository lessonRepository;
    Student student;
    @BeforeEach
    public void init() {
        student = Student.builder().name("Jorn").surname("Deynhoven").build();
        studentRepository.save(student);
        lessonRepository.saveAll(List.of(
                Lesson.builder().name("Music").courseLength(10).student(student).courseLevel(123).build
(),
                Lesson.builder().name("Art").courseLength(20).courseLevel(123).student(student).build(),
                Lesson.builder().name("Trance").courseLength(30).courseLevel(123).student(student).inPro
gress(true).build()
        ));
    }
    @Test
    void deleteStudentWithLessonsTest() throws SQLException {
        studentService.deleteStudentWithLessons(student.getId());
    }
}
```

Interviewer guidelines

Candidate has to know how to use spring data in any spring application which uses spring data. If an entity wanted to remove from database without removing the child attributes, then first relation between them should be removed. Without doing this, spring will throw DataIntegrityViolationException to keep database consistent. To achive that without changing existing code, we need to annotate deleteStudentWithLessons(Long id) method with @Transactional annotation. With this spring will update the database states of child entities. In this case it will update Lesson attributes to have student as null. Then student will be removed in the next line.

After this there is a inProgress check for existing lessons. it will throw SQLException which is a checked exception. @Transactional annotation in default only rolls back for unchecked exception such as RuntimeException. So all changes will be reflected to database.

Option A and E is right. More info can be found in https://docs.spring.io/spring-framework/docs/current/javadoc-api/org/springframework/transaction/annotation/Transactional.html

Candidate's Solution

Options: (Expected answer indicated with a tick)

- [x] ** It will try to delete the student then the DataIntegrityViolationException will be thrown. **

- [ ] It will delete the student successfully then an SQLException will be thrown with the message "Cannot delete all Lessons. There are lessons in progress state".

- [ ] If @Transactional annotation is put in place of X (StudentServer.java), after saving the student and lessons, it will update lessons and remove the student. Then it will throw an SQLException and all database changes will be rolled back.

- [ ] It will throw a RuntimeException with the message "Student not found".

- [ ] If @Transactional annotation is put in place of X (StudentServer.java), after saving the student and lessons, it will update lessons and remove the student. Then it will throw an SQLException.

No comments.

4. Endpoint Security

Correct

Multiple Choice | Spring Boot | Medium

Question description

In Spring Boot Security, the option to enable secure endpoints with @PreAuthorize annotation is:

Interviewer guidelines

@EnableGlobalMethodSecurity by default doesn't enable this feature. To enable this prePostEnabled=true value must be set

Candidate's Solution

Options: (Expected answer indicated with a tick)

- ( ) Configuration class annotated with @EnableGlobalMethodSecurity
- ( ) Configuration class annotated with @EnableGlobalMethodSecurity(securedEnabled = true)

- [x] Configuration class annotated with @EnableGlobalMethodSecurity(prePostEnabled = true)
- [ ] None above

No comments.

5. Predict the Output

Correct

Multiple Choice | Medium | Exception Handling | Spring Boot | Aspect-Oriented Programming

Question description

Consider the following Snippet and predict the output.

The following API is triggered: http://localhost:8080/add/employee?eId=123&fName=Ravi&lastName=Shekhar

```
@Autowired
private EService eService;

@RequestMapping(value = "/add/employee", method = RequestMethod.GET)
public com.model.Employee add(@RequestParam("eId") String eId, @RequestParam("fName") Strin
g fName, @RequestParam("lastName") String lastName)
{
   return eService.createEmployee(eId, fName, lastName);
}

@Aspect
@Component
public class EServiceAspect
{
   @After(value = "execution(* com.random.service.EService.*(..)) && args(eId, fName, lastName)")
   public void afterAdvice(JoinPoint joinPoint, String eId, String fName, String lastName) {
```

```
        System.out.println("After method:" + joinPoint.getSignature());
        System.out.println("Creating Employee with first name - " + fName+ ", second name - " + lastNa
me+ " and id - " + eId);
    }
}
```

Candidate's Solution

Options: (Expected answer indicated with a tick)

- [x] ** After method : com.test.model.Employee addEmployee(String,String,String) Creating Employee with first name - Ravi , second name - Shekhar and id - 123   **

- [ ] After method : com.test.model.Employee addEmployee(string,string,string) Creating Employee with first name - Ravi , second name - Shekhar and id - 123

- [ ] After method : com.test.model.Employee addEmployee(empId,firstName,secondName) Creating Employee with first name - Ravi , second name - Shekhar and id - 123

- [ ] After method : com.test.model.Employee addEmployee(String,String,String) Creating Employee with first name - Shekhar, second name - Ravi and id - 123

No comments.

6. Minimize Path Value

Correct

Coding Graphs Medium

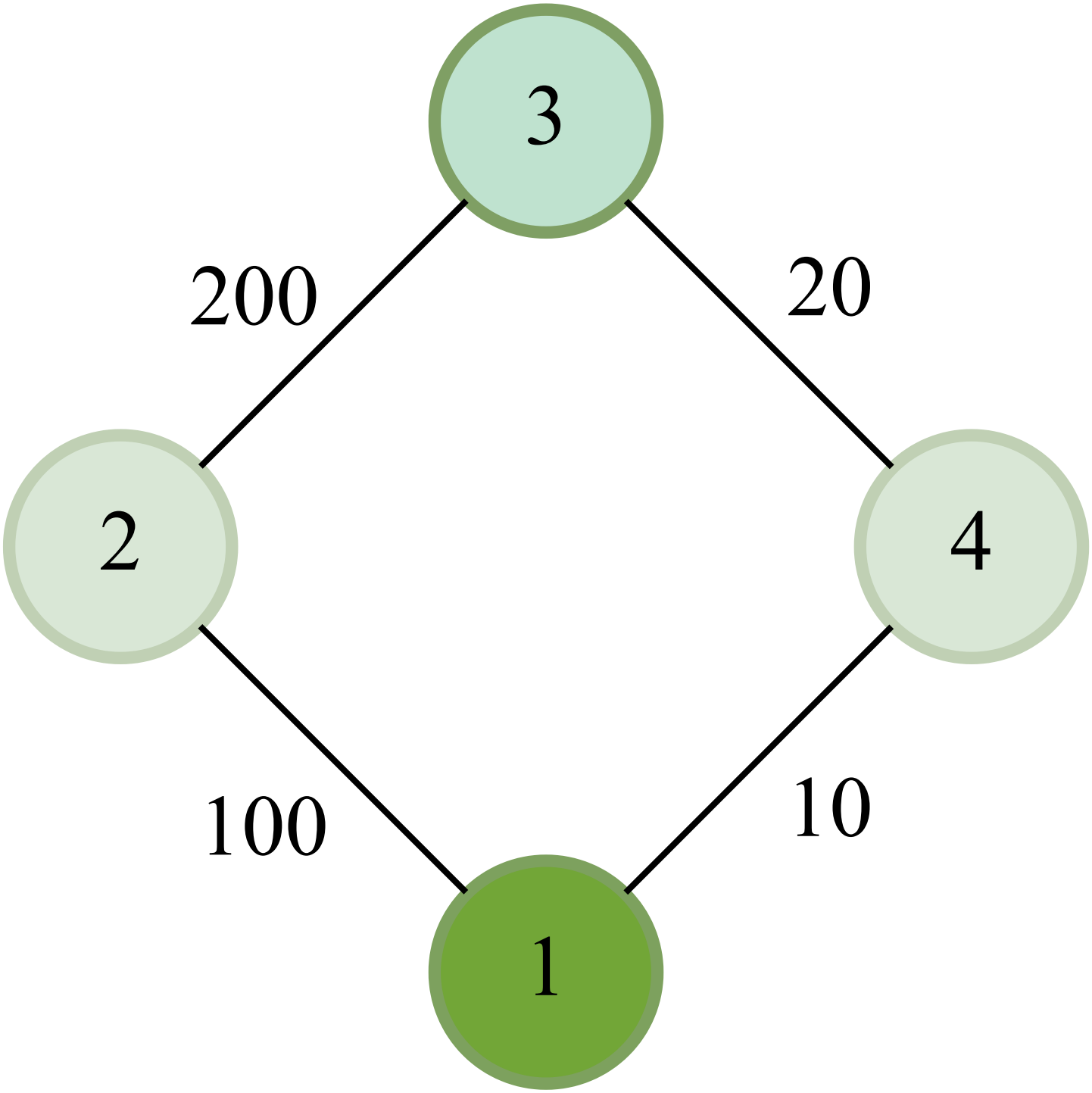
Question description

You have a weighted undirected graph with *graph_nodes* nodes and *m* edges. The stress level of a path between two nodes is defined as the weight of the heaviest edge in that path.

Given a source node *source* and a destination node *destination*, find the minimum possible stress level of a path. If no such path exists, return -1.

Example

Consider the following example below, where *source=1* and *destination=3.*



There are two paths, from node 1 to node 3.

1. The max weighted edge/stress level in path 1 -> 2 -> 3 is 200.
2. The max weighted edge/stress level in the path 1 -> 4 -> 3 is 20.

Return 20, the lower stress level from the second path.

Function Description

Complete the function *getMinimumStress* in the editor with the following parameter(s):

int graph_nodes: number of nodes
int graph_from[m]: one end node for an edge
int graph_to[m]: one end node for an edge
int graph_weight[m]: the weights of the edges
int source: the source node
int destination: the destination node

Returns

int: the value of the least stressful path, or -1 if no path exists

Note: Even though edge endpoints are called *graph_from[i]* and *graph_to[i]*, this is an undirected graph.

Constraints

- $1 \le graph_nodes \le 10^5$
- $1 \le |graph_from| = |graph_to| = |graph_weight| \le 10^5$ *(Where |A| denotes the size of array A)*
- $1 \le graph_from[i], graph_to[i], source, destination \le graph_nodes$
- $0 \le graph_weight[i] \le 10^9$

▼ INPUT FORMAT FOR CUSTOM TESTING

The first line contains two space-separated integers *graph_nodes* and *m,* the number of nodes, and the number of edges of the graph.
Each of the following *m* lines contains three space-separated integers *u, v,* and *w,* meaning there is an edge connecting the nodes *u* and *v* of weight *w.*
The next line contains a single integer, denoting the index of the "*source*" node.
The next line contains a single integer, denoting the index of the "*destination*" node.

▼ SAMPLE CASE 0

Sample Input For Custom Testing

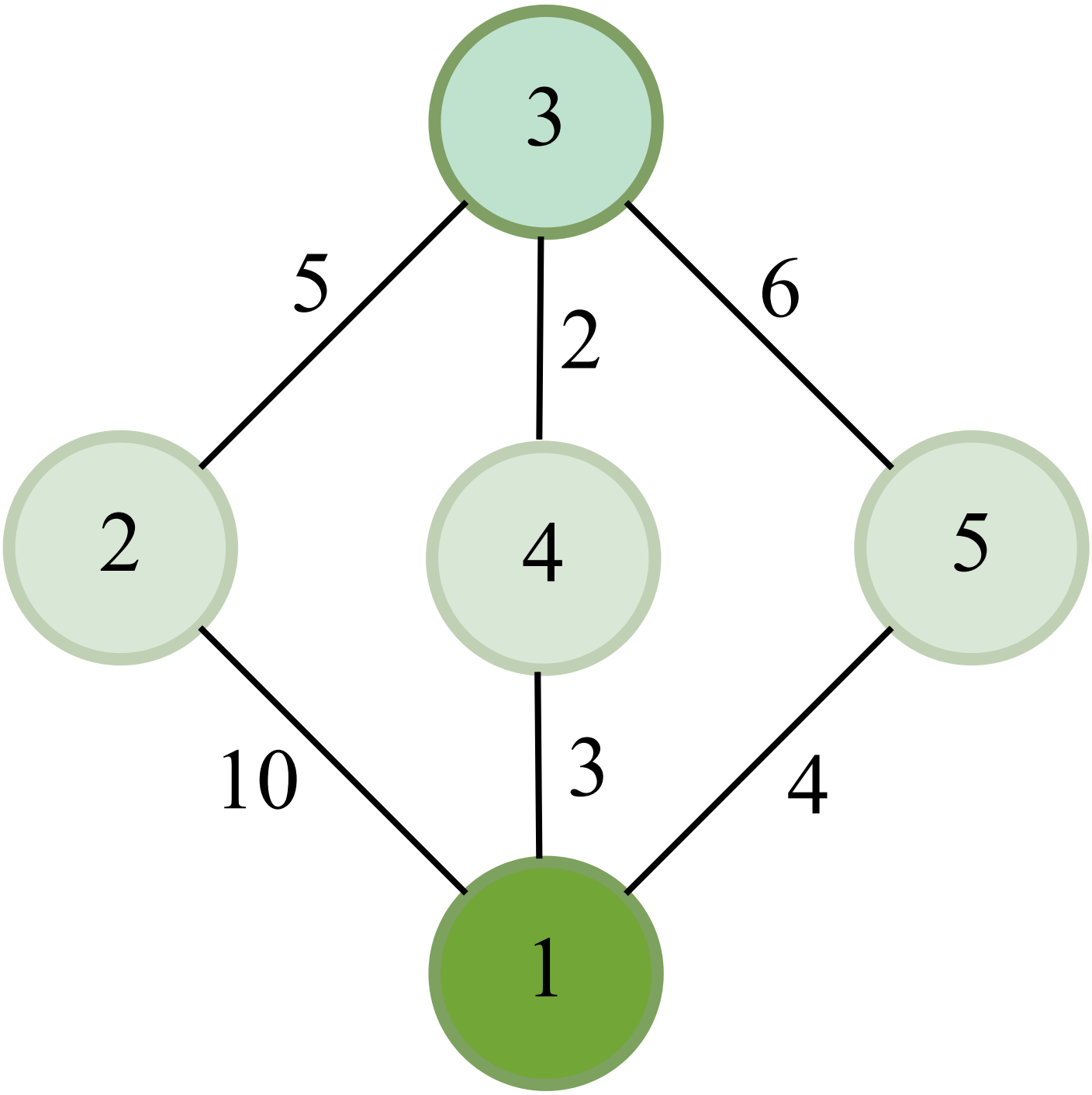
```
STDIN       FUNCTION
-----       --------
5 6         graph_nodes = 5, m = 6
1 2 10      u = 1, v = 2, w = 10
```

```
2 3 5
1 4 3
4 3 2
1 5 4
5 3 6
1          source = 1
3          destination = 3
```

Sample Output

```
3
```

Explanation



There are three paths, from node 1 to node 3 :

1. 1 -> 2 -> 3. Edges have weights 10 and 5, the maximum weighted edge, and the stress level is 10.
2. 1 -> 4 -> 3. Edges have weights 3 and 2, and the stress level = 3.
3. 1 -> 5 -> 3. Edges have weights 4 and 6, and the stress level = 6.

▼ SAMPLE CASE 1

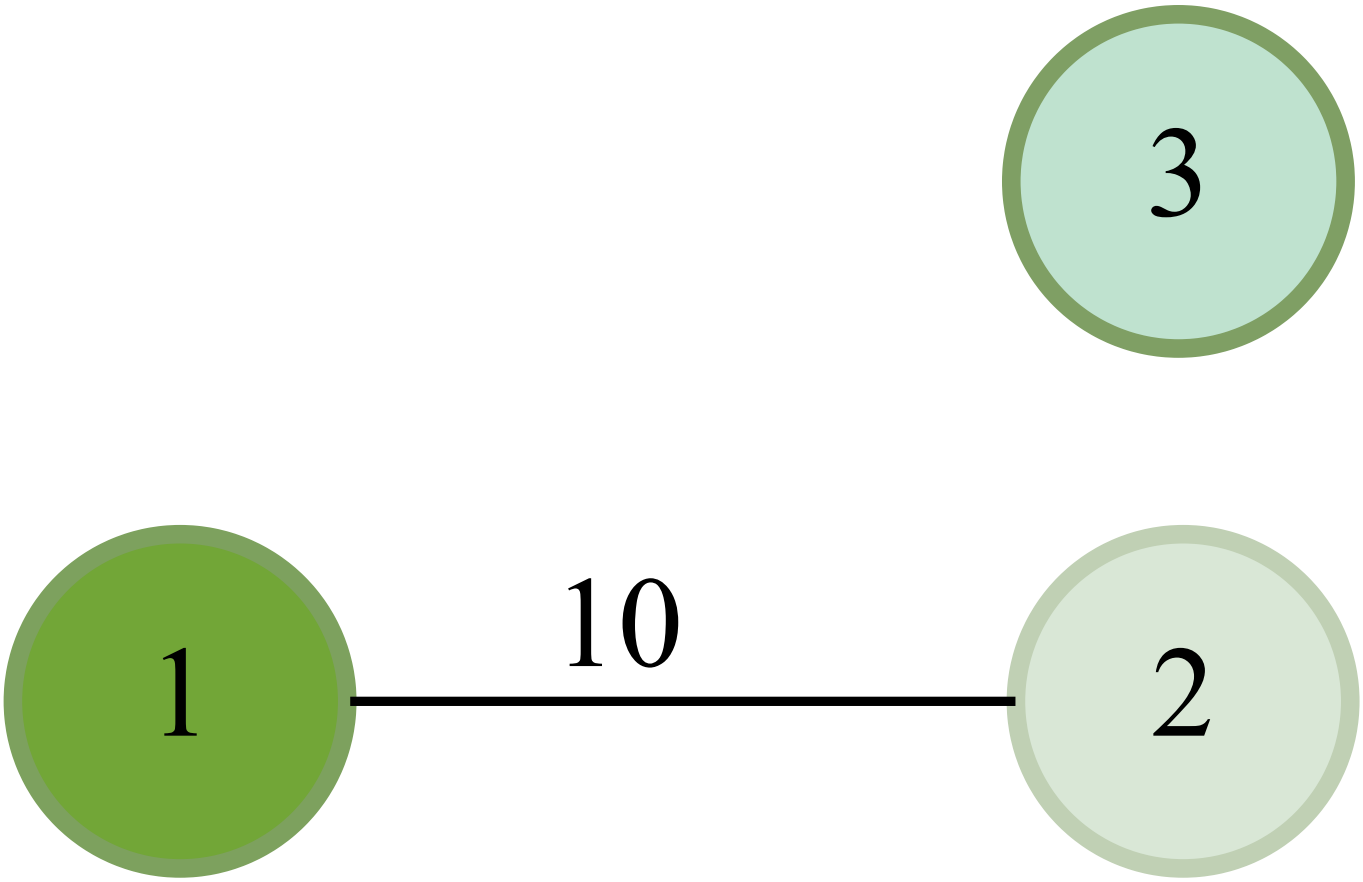
Sample Input For Custom Testing

```
STDIN       FUNCTION
-----       --------
3 1         graph_nodes = 3, m = 1
1 2 10      a = 1, b = 2, w = 10
1           source = 1
3           destination = 3
```

Sample Output

```
-1
```

Explanation



There is no path, from node 1 to node 3.

Interviewer guidelines

▼ SOLUTION

Python Solution

```
sys.setrecursionlimit(10**5)


def getMinimumStress(graph_nodes, graph_from, graph_to, graph_weight, source, destination):

    adj = [[] for i in range(graph_nodes + 1)]
    vis = [False] * (graph_nodes + 1)

    for x, y, w in zip(graph_from, graph_to, graph_weight):
        adj[x].append((y, w))
        adj[y].append((x, w))

    def dfs(node, mx):
        vis[node] = True
        if node == destination:
            return True
        ans = False
        for i, w in adj[node]:
            if w <= mx and not vis[i]:
                ans |= dfs(i, mx)
        return ans

    if not dfs(source, max(graph_weight)):
        return -1

    lo, hi = 0, max(graph_weight)
    while hi - lo > 1:
        vis = [False] * (graph_nodes + 1)
        mid = (hi + lo) // 2
        if dfs(source, mid):
            hi = mid
        else:
            lo = mid
    return hi
```

Candidate's Solution

Language used: **Java 8**

```
import java.io.*;
import java.math.*;
import java.security.*;
import java.text.*;
import java.util.*;
import java.util.concurrent.*;
```

```java
import java.util.function.*;
import java.util.regex.*;
import java.util.stream.*;
import static java.util.stream.Collectors.joining;
import static java.util.stream.Collectors.toList;


class Result {

    /*
     * Complete the 'getMinimumStress' function below.
     *
     * The function is expected to return an INTEGER.
     * The function accepts following parameters:
     *  1. WEIGHTED_INTEGER_GRAPH graph
     *  2. INTEGER source
     *  3. INTEGER destination
     */

    /*
     * For the weighted graph, <name>:
     *
     * 1. The number of nodes is <name>Nodes.
     * 2. The number of edges is <name>Edges.
     * 3. An edge exists between <name>From[i] and <name>To[i]. The weight of the edge is <name>Weight[i].
     *
     */
    static class DSU{
        int[] parent;
        int[] rank;
        DSU(int n){
            parent = new int[n+1];
            rank = new int[n+1];
            for(int i =1;i<=n;i++){
                parent[i] =i;
            }
        }
        int find(int x){
            if(parent[x] !=x)
                parent[x] = find(parent[x]);
                return parent[x];
        }
        void union(int x, int y){
            int px = find(x);
```

```
            int py = find(y);
            if(px!= py){
                if(rank[px] < rank[py]){
                    parent[px] = py;
                } else if(rank[px] > rank[py]){
                    parent[py] = px;
                } else{
                    parent[py] =px;
                    rank[px]++;
                }
            }
        }
    }

    public static int getMinimumStress(int graphNodes, List<Integer> graphFrom, List<Integer> graphTo, List<Integer> graphWeight, int source, int destination) {
            int m = graphFrom.size();
            List<int[]> edges = new ArrayList<>();
            for(int i =0;i<m;i++){
                edges.add(new int[]{
                graphFrom.get(i),
                graphTo.get(i),
                graphWeight.get(i)
                });
            }
            edges.sort(Comparator.comparingInt(a -> a[2]));
            DSU dsu = new DSU(graphNodes);
            for(int[] edge: edges){
                int u = edge[0];
                int v = edge[1];
                int w = edge[2];
                dsu.union(u,v);
                if(dsu.find(source)== dsu.find(destination)){
                    return w;
                }
            }
            return -1;
        }
}
public class Solution {
    public static void main(String[] args) throws IOException {
        BufferedReader bufferedReader = new BufferedReader(new InputStreamReader(System.in));
        BufferedWriter bufferedWriter = new BufferedWriter(new FileWriter(System.getenv("OUTPUT_PATH")));
```

```java

        String[] graphNodesEdges = bufferedReader.readLine().replaceAll("\\s+$", "").split(" ");

        int graphNodes = Integer.parseInt(graphNodesEdges[0]);
        int graphEdges = Integer.parseInt(graphNodesEdges[1]);

        List<Integer> graphFrom = new ArrayList<>();
        List<Integer> graphTo = new ArrayList<>();
        List<Integer> graphWeight = new ArrayList<>();

        IntStream.range(0, graphEdges).forEach(i -> {
            try {
                String[] graphFromToWeight = bufferedReader.readLine().replaceAll("\\s+$", "").split(" ");

                graphFrom.add(Integer.parseInt(graphFromToWeight[0]));
                graphTo.add(Integer.parseInt(graphFromToWeight[1]));
                graphWeight.add(Integer.parseInt(graphFromToWeight[2]));
            } catch (IOException ex) {
                throw new RuntimeException(ex);
            }
        });

        int source = Integer.parseInt(bufferedReader.readLine().trim());

        int destination = Integer.parseInt(bufferedReader.readLine().trim());

        int result = Result.getMinimumStress(graphNodes, graphFrom, graphTo, graphWeight, source, destination);

        bufferedWriter.write(String.valueOf(result));
        bufferedWriter.newLine();

        bufferedReader.close();
        bufferedWriter.close();
    }
}

```

TESTCASE	DIFFICULTY	TYPE	STATUS	SCORE	TIME TAKEN	MEMORY USED

TestCase 0	Easy	Sample	Success	1	0.12 sec	31.8 KB
TestCase 1	Easy	Sample	Success	1	0.1027 sec	31.3 KB
TestCase 2	Easy	Sample	Success	1	0.1207 sec	31.6 KB
TestCase 3	Easy	Hidden	Success	6	0.4571 sec	140 KB
TestCase 4	Easy	Hidden	Success	6	0.5476 sec	133 KB
TestCase 5	Easy	Hidden	Success	6	0.5922 sec	134 KB
TestCase 6	Easy	Hidden	Success	6	0.5302 sec	150 KB
TestCase 7	Easy	Hidden	Success	6	0.4966 sec	154 KB
TestCase 8	Easy	Hidden	Success	6	0.5067 sec	138 KB
TestCase 9	Easy	Hidden	Success	6	0.6051 sec	147 KB
TestCase 10	Easy	Hidden	Success	6	0.5258 sec	147 KB
TestCase 11	Easy	Hidden	Success	6	0.4766 sec	140 KB
TestCase 12	Easy	Hidden	Success	6	0.4206 sec	80.3 KB

TestCase 13	Easy	Hidden	Success	6	0.386 sec	97.1 KB
TestCase 14	Easy	Hidden	Success	6	0.5149 sec	153 KB

No comments.

7. Java: Optimized Sequential Video Watching

Correct

Coding Medium Greedy Java

Question description

A learner is given a sequential playlist of *n* videos to enhance their knowledge. Implement a function to calculate the minimum time required to complete a total of *m* viewings while adhering to the platform's constraints.

System Constraints

- Sequential Viewing:
  - The learner must watch videos in the order they appear in the playlist.
  - To watch a video at index *i*, the learner must first watch all videos from index 0 to *i-1* at least once.
- Initial Full Viewing:
  - To view any video at index *i* for the first time, the learner must spend *firstWatch[i] + repeatWatch[i]* minutes. This ensures a complete understanding of the video.
- Subsequent Viewings:
  - For subsequent viewings of a video at index *i*, the learner spends only *repeatWatch[i]* minutes.
  - At any time, they can rewatch any video they have already watched.
- Minimum Total Watch Count:
  - The learner must achieve at least *m* total viewings across all videos.
  - Videos can be rewatched as desired, and not all videos need to be watched equally.

Return the minimum time required to complete *m* viewings of videos while following these constraints.

Example

If *n* = 4, *m* = 4, *firstWatch* = [1, 5, 9, 11], and *repeatWatch* = [2, 7, 10, 11], the optimal strategy is as follows:

- The learner watches the 0th video for the first time, spending *firstWatch*[0] + *repeatWatch*[0] = 1 + 2 = 3 minutes.
- The learner then rewatches the 0th video 3 more times in, spending *repeatWatch*[0]*3 = 2 * 3 = 6 minutes.
- The learner has successfully completed 4 viewings. Thus, the total time required is 3 + 6 = 9 minutes, and 9 is returned.

Function Description

Complete the function *findMinMinutes* in the editor below.

Function Parameters

int m: the total number of video viewings to complete

int firstWatch[n]: the minutes required to watch each video the first time

int repeatWatch[n]: the minutes required to rewatch each video

Returns

long: the minimum minutes required for the learner to reach their target number of viewings

Constraints

- $1 \le n \le 10^5$
- $1 \le m \le 10^9$
- $1 \le firstWatch[i], repeatWatch[i] \le 10^9$

▼ INPUT FORMAT FOR CUSTOM TESTING

The first line contains an integer, *m*.
The next line contains an integer, *n,* the size of *firstWatch*.
Each of the next *n* lines contains an integer, *firstWatch[i]*.
The next line contains an integer, *n,* the size of *repeatWatch*.
Each of the next *n* lines contains an integer, *repeatWatch[i]*.

▼ SAMPLE CASE 0

Sample Input 0

```
STDIN    FUNCTION
-----    --------
5        m = 5
3        firstWatch[] size n = 3
6        firstWatch = [6, 8, 9]
8
9
3        repeatWatch[] size n = 3
4        repeatWatch = [4, 2, 10]
2
10
```

Sample Output 0

```
26
```

Explanation

The goal is to complete 5 viewings. One optimal strategy is:

- The learner watches the 0th video for the first time, so the time required is *firstWatch*[0] + *repeatWatch*[0] = 6 + 4 = 10 minutes.
- The learner watches the 1st video for the first time, so the time required is *firstWatch*[1] + *repeatWatch*[1] = 8 + 2 = 10 minutes.
- The learner then rewatches the 1st video 3 times. So the time taken = *repeatWatch*[1]*3 = 2 * 3 = 6 minutes.
- The total time required to complete 5 video viewings is 10 + 10 + 6 = 26 minutes.

▼ SAMPLE CASE 1

Sample Input 1

```
STDIN     FUNCTION
-----     --------
6         m = 6
5         firstWatch[] size n = 5
2         firstWatch = [2, 3, 7, 8, 4]
3
7
8
4
5         repeatWatch[] size n = 5
3         repeatWatch = [3, 6, 8, 7, 6]
6
```

```
8
7
6
```

Sample Output 1

```
20
```

Explanation

The goal is to complete 6 viewings. One optimal strategy is:

- The learner watches the 0th video for the first time in *firstWatch*[0] + *repeatWatch[0]* = 2 + 3 = 5 minutes.
- The learner then rewatches the 0th video 5 times in *repeatWatch[0]**5 = 3 * 5 = 15 minutes.
- The total time required to complete 6 viewings is 5 + 15 = 20 minutes.

Interviewer guidelines

▼ SOLUTION

Description

The greedy solution suggests that, at any given index, if the time to watch the video is less than at any other index between i+1 and n, the minimum total time will be the sum of the watch time up to index i-1, plus the repeat time of the ith video multiplied by (m-i).

Solution

```
class Result {

    /*
     * Complete the 'findMinMinutes' function below.
     *
     * The function is expected to return a LONG_INTEGER.
     * The function accepts following parameters:
     *  1. INTEGER m
     *  2. INTEGER_ARRAY firstWatch
     *  3. INTEGER_ARRAY repeatWatch
     */

    public static long findMinMinutes(int m, List<Integer> firstWatch, List<Integer> repeatWatch) {
        long minMinutes = Long.MAX_VALUE;
        long cur = 0;
        int n = firstWatch.size();

        for (int i = 0; i < Math.min(n,m); i++) {
```

```
            int a = firstWatch.get(i);
            int b = repeatWatch.get(i);

            minMinutes = Math.min(minMinutes, cur + ((long) (m - i) * b) + a);

            cur += a + b;
        }
        return minMinutes;
    }

}
```

▼ COMPLEXITY

Time Complexity - O(n)

The algorithm iterates through each number in *firstWatch* and *repeatWatch* exactly once, performing constant-time checks and arithmetic operations in each iteration. Thus, the time complexity is O(n), where *n* is the length of *firstWatch*.

Space Complexity - O(1)

Space usage is constant, only utilizing a fixed number of variables (*mn, cur,* etc.) that do not scale with the length of *firstWatch*. Therefore, the space complexity is O(1).

Candidate's Solution

Language used: **Java 8**

```java
import java.io.*;
import java.math.*;
import java.security.*;
import java.text.*;
import java.util.*;
import java.util.concurrent.*;
import java.util.function.*;
import java.util.regex.*;
import java.util.stream.*;
import static java.util.stream.Collectors.joining;
import static java.util.stream.Collectors.toList;


class Result {

    /*
```

```
     * Complete the 'findMinMinutes' function below.
     *
     * The function is expected to return a LONG_INTEGER.
     * The function accepts following parameters:
     *  1. INTEGER m
     *  2. INTEGER_ARRAY firstWatch
     *  3. INTEGER_ARRAY repeatWatch
     */

    public static long findMinMinutes(int m, List<Integer> firstWatch, List<Integer> repeatWatch) {
            int n = firstWatch.size();
            long totalTime = 0;
            long totalViews =0;
            long minRepeat = Long.MAX_VALUE;
            long answer = Long.MAX_VALUE;

            for(int i =0; i<n;i++){
                totalTime += (long) firstWatch.get(i) +repeatWatch.get(i);
                totalViews++;
                minRepeat = Math.min(minRepeat, repeatWatch.get(i));
                if(totalViews>=m){
                    answer = Math.min(answer, totalTime);
                }else{
                    long remaining = m-totalViews;
                    long candidate = totalTime +remaining*minRepeat;
                    answer = Math.min(answer, candidate);
                }
            }return answer;
    }

}

public class Solution {
    public static void main(String[] args) throws IOException {
        BufferedReader bufferedReader = new BufferedReader(new InputStreamReader(System.in));
        BufferedWriter bufferedWriter = new BufferedWriter(new FileWriter(System.getenv("OUTPUT_PATH")));

        int m = Integer.parseInt(bufferedReader.readLine().trim());

        int firstWatchCount = Integer.parseInt(bufferedReader.readLine().trim());

```

```
        List<Integer> firstWatch = IntStream.range(0, firstWatchCount).mapToObj(i -> {
            try {
                return bufferedReader.readLine().replaceAll("\\s+$", "");
            } catch (IOException ex) {
                throw new RuntimeException(ex);
            }
        })
            .map(String::trim)
            .map(Integer::parseInt)
            .collect(toList());

        int repeatWatchCount = Integer.parseInt(bufferedReader.readLine().trim());

        List<Integer> repeatWatch = IntStream.range(0, repeatWatchCount).mapToObj(i -> {
            try {
                return bufferedReader.readLine().replaceAll("\\s+$", "");
            } catch (IOException ex) {
                throw new RuntimeException(ex);
            }
        })
            .map(String::trim)
            .map(Integer::parseInt)
            .collect(toList());

        long result = Result.findMinMinutes(m, firstWatch, repeatWatch);

        bufferedWriter.write(String.valueOf(result));
        bufferedWriter.newLine();

        bufferedReader.close();
        bufferedWriter.close();
    }
}

```

TESTCASE	DIFFICULTY	TYPE	STATUS	SCORE	TIME TAKEN	MEMORY USED
Testcase 2	Easy	Sample	Success	1	0.1271 sec	31.8 KB

Testcase 0	Easy	Sample	Success	1	0.1132 sec	31.9 KB
Testcase 1	Easy	Sample	Success	1	0.1102 sec	31.6 KB
Testcase 3	Easy	Hidden	Success	4	0.1087 sec	31.6 KB
Testcase 4	Easy	Hidden	Success	4	0.1141 sec	32.1 KB
Testcase 5	Easy	Hidden	Success	4	0.197 sec	36.4 KB
Testcase 6	Medium	Hidden	Success	5	0.3551 sec	88.7 KB
Testcase 7	Medium	Hidden	Success	5	0.5451 sec	175 KB
Testcase 8	Medium	Hidden	Success	5	0.5558 sec	175 KB
Testcase 9	Medium	Hidden	Success	5	0.5196 sec	175 KB
Testcase 10	Medium	Hidden	Success	5	0.5274 sec	175 KB
Testcase 11	Medium	Hidden	Success	5	0.4199 sec	164 KB
Testcase 12	Hard	Hidden	Success	10	0.3419 sec	94.4 KB
Testcase 13	Hard	Hidden	Success	10	0.4358 sec	164 KB

Testcase 14	Hard	Hidden	Success	10	0.1141 sec	32 KB

No comments.